

SAIKAT DAS

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📍 Varanasi, India

EDUCATION

Indian Institute of Technology (BHU), Varanasi
Bachelor of Technology in Mining Engineering; CGPA: 8.39/10.0

Varanasi, India
Jul 2023 – May 2027

TECHNICAL SKILLS

Languages & Frameworks	Go, Python, C++ , Dart, FastAPI, Flutter, React
Backend & Distributed Sys	REST APIs, gRPC, SSE, Webhooks, Microservices, Kafka, Redis Async
Agentic AI & LLMs	Google ADK, OpenRouter, Context Engineering, Guard Models, Hallucination Mitigation
Databases & Infra	PostgreSQL, Redis, Apache Solr, Memgraph, Docker, Linux, Git, Grafana k6

EXPERIENCE

Software Engineering Intern | *COPIOUS · Tripple Double* Jun 2026 – Aug 2026

- Built a **Flutter** split-screen web portal with auto-save and rule-based validation for FIBA scoresheet digitization.
- Engineered a **VLM** OCR pipeline via **OpenRouter**, A/B testing models to optimize accuracy vs. cost.
- Reduced hallucination and token costs via dynamic prompts, **JSON key aliasing**, and output-embedded reasoning.
- Cut operator cost by **91.92%** (\$0.52 → \$0.042) and entry time by **73.33%** (30m → 8m) via pre-filled entry.

Flutter Developer Intern | *Vrajpath iTech LLP* Dec 2024 – Jan 2025

- Reduced API requests by ~30% via **Riverpod** state caching, validated in **Dart DevTools**.
- Integrated **Razorpay** for secure payments and **Google Maps** for real-time geolocation.

PROJECTS

Flowback – AI Revenue Recovery Platform | *Go, React, Redis, Async, Google ADK* | 🐙 [dis70rt/Flowback](#) | 📺 [Demo](#)

An autonomous, event-driven pipeline that intercepts failed SaaS payments and deploys AI agents to recover revenue.

- Queued webhooks via **Go + Redis Async**, scaling workers freely with idempotent, effectively-once processing.
- Orchestrated a multi-agent **Google ADK** graph where copywriter and **Gemini TTS** agents drive outreach.
- Wrote a **Go** policy engine enforcing hard rules (discount & contact limits) so no hallucination touches a transaction.
- Streamed agent state to a **HITL React** dashboard via **Redis Pub/Sub** and **SSE**, eliminating all polling overhead.

TradeOrders – Order Matching Engine | *Go, Kafka, PostgreSQL, Redis, Docker* | 🐙 [dis70rt/TradeOrders](#)

Solved lock contention in high-volume trading via an event-driven architecture, scaling throughput 18x.

- Decoupled API from DB via **Kafka**, eliminating all I/O bottlenecks; scaled throughput to **18.8k req/s** (0 errors).
- Partitioned Kafka topics by instrument for **lock-free**, sequential matching, enabling concurrent process per symbol.
- **Grafana k6** load tested at **7,624 VUs**: hit **18.2k req/s** sustained throughput with a **231ms p95 latency**.

Bluppi – Shared Music Playback | *Go, gRPC, PostgreSQL, Redis, Solr, Flutter* | 🐙 [backend](#) / [frontend](#) | 🌐 [bluppi.saikat.in](#)

A real-time synchronization engine that fixes network lag across distributed devices.

- Designed **PTP-style** clock synchronization with **quorum-based** buffering for consistent playback across devices.
- Tracks online presence via a **gRPC** gateway: **51k** concurrent connections with 0 errors and **p99 15ms** latency.
- Live ephemeral chat via **Redis Pub/Sub**, social features via **Memgraph**, and a 500GB music catalog via **Solr**.

QQL (Quick Query Language) – Approximate Query Processing Shell | *Python, PostgreSQL* | 🐙 [dis70rt/qql](#)

A shell sitting above PostgreSQL that accelerates massive dataset aggregations 10x using bounded statistical error.

- Designed a dynamic **pilot query** engine to estimate statistical sample sizes, supporting **COUNT**, **SUM**, **AVG** functions.
- Benchmarked against exact PostgreSQL queries: achieved a **10x speedup** at 5% error and **6x** at 1% error bound.
- Awarded **Special Mention** at the **e6Data Hackathon 2024** for innovative approaches to database acceleration.

ACHIEVEMENTS

Won 1st Place at the BNY Hackathon , building a solution for transaction processing & settlement .	Apr 2026
Received a Special Mention at the e6Data Hackathon for building a QQL system.	Oct 2025
Secured 2nd Place at the Nutanix Hackathon by building WizFlow , a real-time workflow engine .	Apr 2025
Ranked in the Top 9.22% globally on LeetCode with a 1,778 peak rating and 256+ problems solved.	Present
Achieved Pupil rank (1,217 Max Rating) on Codeforces , solving 150+ algorithmic challenges.	Present